Mini-Challenge 1 Report

Visual Analytics for Big Data

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Group 3

Introduction

In recent years, Oceanus Folk has evolved from a local genre that was developed in the island nation of Oceanus, into a widely recognized one, beyond its familiar origins. Sailor Shift is the artist responsible for this major development, particularly her work reaching global audiences. Her early career was strongly influenced by Oceanus Folk, and she still continues to engage with this genre, even though her musical style has expanded further.

One of the first chapters of Sailor's career growth, includes her joining the all-female Oceanus folk band, totaling five members. The band stayed together for three years, leading up to their breaking apart in 2026, due to the artists wishing to each follow their separate individual career paths. Since then, Sailor pursued a solo career, accomplishing a major achievement by 2028, when one of her singles reached the top amongst global charts. Moving on, she has been releasing music consistently, with each song performing better each time, and collaborating with diverse artists, such as Indie Pop and Indie Folk.

Her achievements and success have led to more attention being drawn to Oceanus Folk as a genre, as well as artists associated with it. Her former bandmates have also remained active in the music industry, further confirming the rising influence of Oceanus Folk to the music industry. Consequently, a new generation of artists, under the inspiration of Oceanus Folk, has rose.

This report aims to create visualizations containing information on artists, albums, collaborations, performances and influences to analyze these trends. Therefore, we aim to explore Sailor's career trajectory, identify patterns of collaboration and influence, and dive into how Oceanus Folk evolved over time, from a small local genre, to a global influence.

Literature Review

Data visualization, used effectively, can facilitate data exploration, insights through analysis, and decision making in various domains. The data visualization task at hand utilizes a knowledge graph — a structured representation of human knowledge that consists of entities, and the relationships that exist between these entities [1]. Our task is more specifically about visually exploring the connections between songs, artists, groups, etc. of the Oceanus music industry.

Looking more closely at visualization tools, prior works have demonstrated the value of separate global and local exploration graphs, where an overview of the insights allows users to explore the data in its entire perspective, before inspecting the details [3]. This laid the foundation for part of the Q1 network visualizations, where the concentric ring layout provides an overview of Sailor's collaborations and direct influences, before the dropdown menu reveals the indirect connections extending from each influence artist. Supporting this design choice, research on knowledge graph interfaces has found that network representations allow users to extract concrete relational insights directly from the graph structure [4], of which also informed the decision to use a network graph in Q1 to visualize the influence and collaboration network surrounding Sailor Shift.

Shneiderman's Visual Information Seeking Mantra ('overview first, zoom and filter, then details-on-demand') provides a foundational principle for the interaction design of our dashboard [5]. This is reflected across all three questions: the streamgraph in Q1 provides a temporal overview of Sailor's influences, the brush selection filters to a specific time period, and the linked bar chart delivers artist-level detail for that selection. Similarly, the time chart in Q2 gives an overview of how Oceanus Folk spread before genre and artist breakdowns are explored in the heatmap and lollipop chart.

Research on ranked-list visualizations show that horizontal bar charts often provide the highest accuracy for ranking and comparison tasks [2]. This influenced our decisions in using ranked bar charts in Q1 and Q3 visualizations, where a relative comparison between artists was the primary goal. Heatmap visualizations have similarly been shown to be effective for reducing visual clutter when displaying multivariate categorical data in a single display space [6], which guided our use of a heatmap in Q2 to summarize genre-level influence patterns across the Oceanus Folk network.

Finally, the use of temporal visualization to track and compare career trajectories, as used in Q3, has been prevalent in the academic field. Systems such as CareerMap [7] have demonstrated the value of timeline-based views for understanding when individuals became influential and how their output evolved over time — an approach we adapt here to compare the career arcs of musical artists. Together, these works provide the theoretical grounding for the visualization and interaction design decisions made across this dashboard, connecting established principles in information visualization to the specific analytical questions posed by the Oceanus music network.

Tasks & Questions

Question 1

Sailor Shift's influence on Oceanus Folk, as well as the broader music industry, is a particular point of interest for Silas Reed, an Oceanus journalist. This first task focuses on understanding Sailor's influence and impact, as well as the trajectory of her career over time. Silas has collected a large dataset with information on musical artists, producers, albums, songs, and influences — making a comprehensive knowledge graph. The structure of this data is presented through nodes and edges, where nodes represent entities (*Person*, *MusicalGroup*, *Song*, *RecordLabel*, *Album*) and edges represent the relationships between them. The goal of this first task is to answer the following questions:

a. Who has she been most influenced by over time?

b. Who has she collaborated with and directly or indirectly influenced?

c. How has she influenced collaborators of the broader Oceanus Folk community?

Before diving into preprocessing the data, the first step in this analysis is to define the semantics of the edge types. This was done by dividing them into four different category sets: `authorship_types` (*ComposerOf*, *PerformerOf*, *LyricistOf*, *ProducerOf*) which maps a work to its credited artist, `influence_types` (*InStyleOf*, *InterpolatesFrom*, *CoverOf*, *DirectlySamples*, *LyricalReferenceTo*) which captures one work referencing or being derived from another, `person_types` (*Person*, *MusicalGroup*) which filters out non-human entities (e.g., RecordLabels) when building artist lookups, and `work_types` (*Songs*, *Albums*) which filters out non-work entities when identifying authorship. The distinction between authorship and influence is crucial here. Authorship edges link an artist to a work they created, answering the question 'what artist made this work?'. Whereas influence edges link one work (or entity) to another in a dependent relationship, answering the question 'which work references another work?'.

After this distinction was made, the next step was to identify the node belonging to Sailor Shift and her group, Ivy Echoes. Once this was identified, two functions then extract her entire catalogue of songs and albums: `get_sailor_songs` and `get_sailor_albums`. Because Sailor performed both as a solo artist and as a member of Ivy Echoes, treating only her solo work would give an incomplete picture of her influences and collaborations. Therefore, `sailor_work_ids` and `sailor_entity_ids` both include Ivy Echoes, ensuring that influence edges connected to any of their output are also credited to Sailor, and that group works are not misclassified as external influences or collaborators. From there, `build_influence_edges` isolates all influence-type edges connecting to and from her producing two directional subsets: `to_sailor`, capturing what works influenced Sailor's works, and `from_sailor`, capturing what her songs have gone on to influence. However, the data structure makes answering 'who influenced who?' less straightforward. Influence edges in this dataset connect songs and albums to other songs and albums, not artists directly (with the exception of *InStyleOf*). An additional resolution step is therefore needed to link each work back to the artist behind it. This is handled through an authorship lookup table: a mapping of (*artist_id*, *work_id*) pairs derived from `authorship_type` edges where the source is a *Person* or *MusicalGroup*. Additionally, genre is attached to each row by the resolved *work_id* through a genre lookup dictionary built from `df_all_nodes`, capturing the genre of the work carrying the influence relationship. Now with the influence edges resolved to their artists, the resulting dataframe, `df_influence_artists`, forms the foundation for answering Silas's first question: Who has she been most influenced by over time?

However, to fully understand Sailor's influence on the music industry, we must go deeper. Silas wants to understand who she has collaborated with and directly or indirectly influenced. Right now, `df_influence_artists` captures who has directly referenced Sailor's work, but tracing indirect influence requires going one step further and identifying the artists influenced by those who were influenced by her. This is handled by constructing a two-hop influence table. Hop1 is drawn directly from `df_influence_artists`: the set of artists whose works reference Sailor. Hop2 is computed through a work-level traversal in four steps: each hop1 artist's full body of works are collected (`artist_to_works` dictionary); all `influence_type` edges originating from those works are identified (`h2_from_raw`); for any edge pointing to a work rather than a person, the authoring artist is resolved via the authorship lookup (these are the hop2 artists); and finally, any artist already present in hop1 or in `sailor_entity_ids` is excluded to prevent circular attribution. Then, a bridge table is created to identify indirect connections in Sailor's influence network by tracing paths from hop1 artists (those directly influenced by Sailor) through their works to any secondary artists (hop2). This allows us to trace the ripple effect of Sailor's influence — artists who were not directly connected to her but are linked through the works of those who were. The second part of this question concerns who Sailor has collaborated with. The function `sailor_collaborations` identifies all artists sharing an authorship credit on any of Sailor's songs or albums by filtering `authorship_types` edges that target `sailor_work_ids`. The output, `df_collaborators`, links each collaborator to the specific work they contributed to. This then ensures that the data is fully set up for visually analyzing the question: Who has she collaborated with and directly or indirectly influenced?

The third question concerns Sailor's broader influence on the Oceanus Folk genre as a whole. Answering this first requires establishing who that community consists of and how active its members have been over time. The function `build_oceanus_community` does this by identifying all songs and albums in `df_all_nodes` where the genre is Oceanus Folk, then resolving their authorship through `authorship_type` edges filtered to `person_type` sources. Each artist is then merged with work metadata to extract release years and deduplicated to one row per (*artist*, *work*) pair, before being aggregated into a per-year count of Oceanus Folk works per artist. The final step flags each artist with `influenced_by_sailor` where their ID appears with a `from_sailor` direction. This then forms the foundation for answering the question: How has she influenced collaborators of the broader Oceanus Folk community?

To answer the question who influenced Sailor most over time, two connecting visualizations are used: a streamgraph tracking genre-level influence across time, and a ranked bar chart identifying the top influencing artists within a selected period. The streamgraph (Figure 1) is built from `sg_data`, a subset of `df_influence_artists` filtered to `to_sailor` edges (to measure who influenced Sailor) and aggregated to (*year*, *genre*) counts. There were several decisions that influenced the design of the streamgraph. For one, a centered baseline (`stack = 'center'`) is used, rather than a zero or normalized, because it allows for independently readable stream shapes. This is appropriate in this case because the question concerns the composition of influence over time, rather than just a running total. Additionally, cubic interpolation (`interpolation = 'monotone'`) was used to smooth the curves between yearly counts. Genre colors are also assigned here by alphabetical order of the genre names to ensure that the colors stay consistent on reruns and between the streamgraph and bar chart. As an added interactive aspect of the visualizations, an `alt.selection_interval` brush on the x-axis is registered as a Marimo `ui.altair_chart` parameter. Its `.value` property is read in the downstream bar chart cell; meaning when a year-range selection is made on the streamgraph through the brush filter, the bar chart is then filtered for those years (when no selection exists, the bar chart defaults to the full year range). The bar chart then focuses on the ranking of artists by influence score (from highest to lowest): (5) *DirectlySamples*, (4) *CoverOf*, (3) *InterpolatesFrom*, (2) *LyricalReferenceTo*, (1) *InStyleOf*. An initial approach to mapping influence to Sailor over time was based on how many works an artist had as influencing Sailor. However, after taking a closer look at Sailor's influence edges, it was discovered that each artist has at most one influence edge towards Sailor, meaning there would be no meaningful differentiation between influence artists. Therefore, this ranking is based on the type of influence towards Sailor, on the premise that different influence edge types represent varying degrees of influence. Directly sampling another artist's work is a more concrete act of influence compared to creating a work in-style of another artist. Therefore, the bar chart then shows the top 10 influence artists based on this ranking, for the selected time period chosen in the streamgraph. Each bar is then colored by the genre of the influence artists' work, using the same color attribution as the streamgraph.

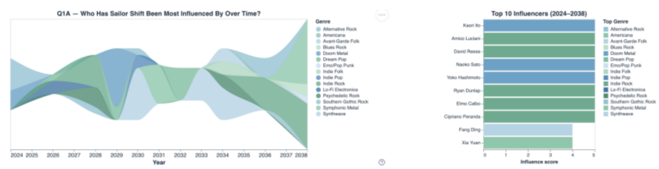

Figure 1: Streamgraph of genre-level influence on Sailor Shift over time (left), with a ranked bar chart of the top 10 influencing artists for the selected period (right).

The streamgraph and bar chart are built from 97 `to_sailor` influence artists that span across her entire career, from 2024 to 2038. Figure 1 shows that Sailor has been influenced by a variety of genres over time, with no clear pattern emerging. Americana is her favorite genre to take inspiration from in 2038, but in 2032 Dream Pop was her sole interest. Overall, Synthwave accounts for 18.6% of all influence rows and Indie Rock 17.5%, together making up over a third (36.1%) of the total. The number of artists she takes influence from does vary as well, with the fewest influence artists at 3 in 2026, but the biggest being in 2038 with 20 artists. This dip in 2026 aligns with when Sailor's group, Ivy Echoes broke up to pursue their individual careers. Sailor's influence from other artists grew with time after this, and there is a slight peak in the number of influence artists in 2028, right when her first single goes viral. The influence score on the right gives a clue as to who Sailor has gained the most inspiration from, based on how direct the influence was. Meaning, Sailor has directly sampled from the top 8 artists shown in Figure 1. However, *InStyleOf* is the most common edge type at 35.1% of rows. The slider used over the streamgraph can then narrow this influence for a specific time period. Figure 2 shows that from 2034 to 2036, Sailor's influence mostly came from covering Simone Säuberlich and Belinda Knappe's works, both Psychedelic Rock.

When looking at who Sailor collaborated with and directly or indirectly influenced, a three-tier network structure

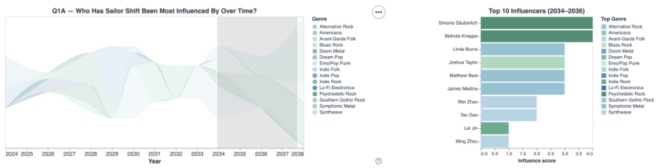

Figure 2: Streamgraph filtered to 2034–2036.

(Figure 3) is used to visualize Sailor's music relationships. Sailor, at the center, all collaborators and directly influenced artists (hop1) in the inner ring, and those artists (hop2) who influenced by the hop1 artists. The hop2 artist expansion is controlled by a drop-down menu with all the hop1 artist names. This was a readability design choice due to the large number of nodes that would be exposed otherwise. The chart also includes zoom and pan options, as the graph can become populated with many nodes.

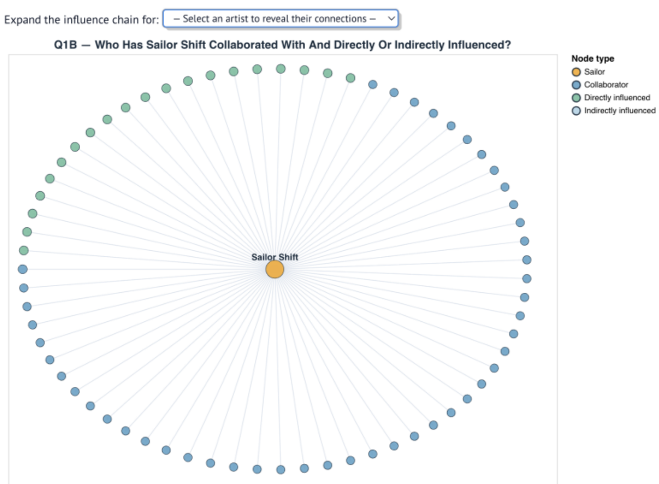

Figure 3: Three-tier network graph.

Sailor has 47 collaborators that are largely part of the Oceanus Folk genre, with 68% of them having it as their primary genre. Maya Jensen, Sailor's former bandmate, is the artist who has most collaborated with Sailor, with 4 shared works. There are also 20 unique artists who have been directly influenced by Sailor, with Julia Carter having the largest influence chain with 55 hop2 artists. However, one thing to note is that there are 3 hop1 artists with no works that carry outgoing influence: N.V. Blake, Ivy Steele, and Scarlett Moon. This means that these 3 artists went on to have no influence on other Oceanus artists. The extended map, as shown for The Hollow Monarchs in Figure 4, shows the chain of influence from Sailor, to the artist she influenced, to then the artists that went on to be influenced, marking the full direct and indirect impact of Sailor's work.

Addressing how Sailor Shift affected the broader Oceanus Folk (OF) community is shown through a stacked area chart (Figure 5) to show the trajectory of OF works across

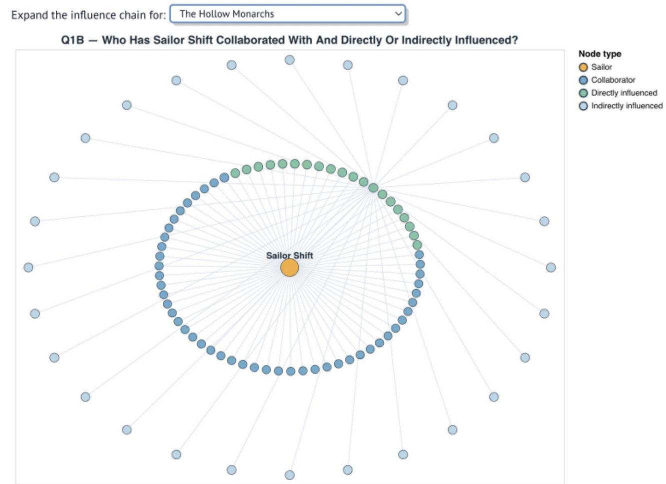

Figure 4: Extended influence map for The Hollow Monarchs, illustrating the full chain of direct and indirect impact originating from Sailor Shift.

the full timeline. The OF community is comprised of 762 artists collapsed into three groups (in the stacking order from bottom-up): no influence, collaborator of Sailor, and influenced by Sailor. A zero baseline is chosen here, in contrast to question 1's streamgraph, because the total height of the chart at any given year represents the total number of OF works released for that year.

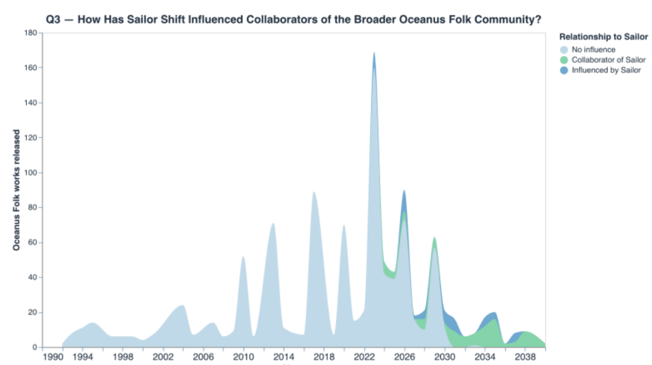

Figure 5: Stacked area chart of Oceanus Folk works over time, split into three groups: no influence, collaborator of Sailor, and influenced by Sailor.

However, there is a key piece of information to point out. The `influenced_by_sailor` and `collaborator` flags only reflect the present relationship to Sailor, and not the date when that relationship was established. This means that a year guard needed to be put into place at 2023 to mark when Sailor started her music career and ensure that influence or collaborators are not shown in a period before she was active. The chart shows that the OF community had a solid base before Sailor Shift entered into the industry, with works spiking at 169 in 2023. In general, collaborators of Sailor's seem to be more visible in the OF community than directly influenced artists, with 40 of the 47 collaborators having worked in the OF genre, but only 5 of the 20 influenced artists being from OF. However, Sailor's influence does spread in the years following her debut, with most works from OF in 2030 and beyond being influenced by or a collaborator of Sailor's.

Question 2

Develop visualizations that illustrate how the influence of Oceanus Folk has spread through the musical world.

a. Was this influence intermittent or did it have a gradual rise?

b. What genres and top artists have been most influenced by Oceanus Folk?

c. On the converse, how has Oceanus Folk changed with the rise of Sailor Shift? From which genres does it draw most of its contemporary inspiration?

In order to analyze the spread of Oceanus Folk, we first filtered songs labeled as Oceanus Folk. We then extracted songs influenced by these tracks using relationship fields, such as 'in-style-of', 'interpolates-from', 'cover-of', 'directly-samples' and lyrical references as well. From these influenced songs, we derived release years, genres, and performing artists to create time-based, genre-based, as well as artist-based visual summaries, utilizing Altair for interactive visualization.

To begin with, as visible in the time chart of Figure 6, the spread of Oceanus Folk shows a gradual rise, rather than intermittent bursts. The time trend visualization was constructed by first aggregating the number of influenced songs per year from the dataset. Then, multiple visual layers were combined, including an area chart to emphasize overall volume, a line and a point layer to show the yearly changes in a detailed way. Finally, a rolling average line, based on a three year window, was added to smooth short term fluctuations. Additionally, a subtle reference horizontal dashed line was added to display the mean level of influence. The latter is very useful to identify below or above average periods.

The influence fluctuates over time, however it does not return to its consistent low levels that were observed up until 2005. There is an overall upward trend, hinting towards a gradual increase with periodic peaks, instead of pure intermittent bursts. The genre experienced a strong acceleration after 2015, with one of the highest achieved peaks being in 2024. This pattern matches sustained growth which was likely driven by the global exposure of Oceanus Folk, thanks to Sailor's talent.

Moving on, according to the heatmap of Figure 7, Oceanus Folk has most strongly influenced Indie Folk and itself, as well as Desert Rock, Synthwave, and Dream Pop, amongst

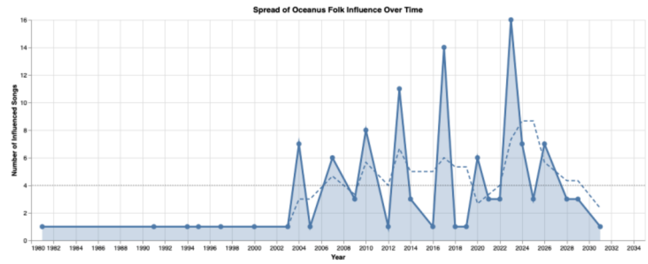

Figure 6: Time trend chart of Oceanus Folk influence over time, including a rolling 3-year average and a mean reference line.

these. More specifically, the genre reinforces itself, with Oceanus Folk influencing new works within its own category. This heatmap was created by aggregating the number of influenced songs by both genre and year, which led to the two-dimensional view of how influence spreads across genres over time. Each cell shows the intensity of the influence, with a darker shade showing a higher influence.

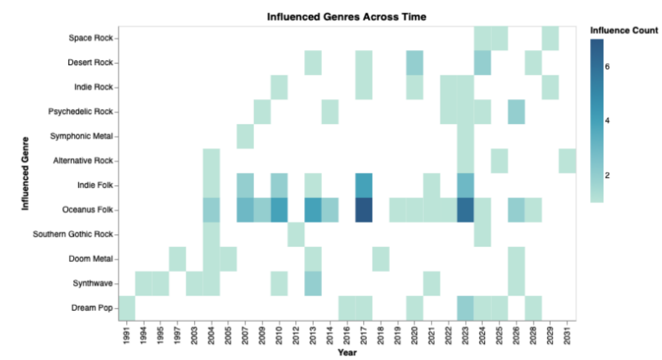

Figure 7: Heatmap showing the intensity of Oceanus Folk's influence across genres and over time (darker = higher influence count).

All in all, based on the lollipop chart of Figure 8, the influence of Oceanus Folk is distributed amongst a wide range of artists, rather than being concentrated on one single individual. To create this chart, we extracted the individual artist names from the `performed_by_persons` field, because multiple artists could be listed in just one record. After cleaning these entries, the data was aggregated to count how often each artist appeared in the influenced songs, and then they were ranked by frequency. Finally, the top 15 were used for the visualization. The reason we selected this type of chart, is because the point at the end of each line reflects the magnitude of the influence, which is more easily readable. Artists such as Stephen Meyer, Alfred Thibault, Lei Jin, Yong Shen show the highest levels of influence. Moving on, several other individuals exhibit moderate influence, all of which shows that Oceanus Folk has contributed to a generation of very diverse emerging artists.

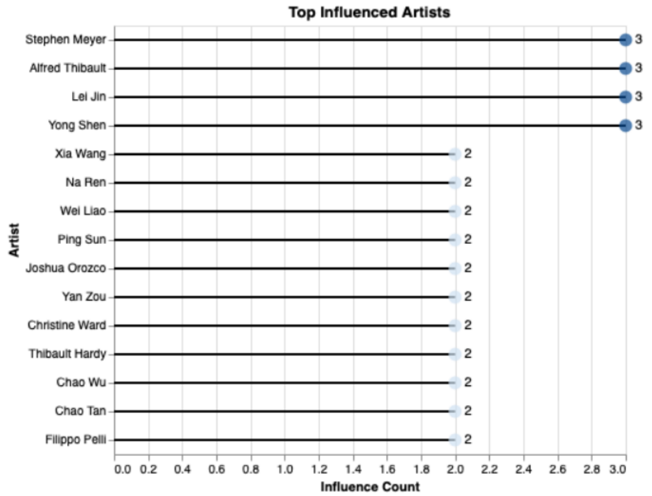

Figure 8: Lollipop chart of the top 15 artists most influenced by Oceanus Folk, ranked by frequency of appearance in influenced songs.

Question 3

Use your visualizations to develop a profile of what it means to be a rising star in the music industry.

a. Visualize the careers of three artists. Compare and contrast their rise in popularity and influence.

b. Using this characterization, give three predictions of who the next Oceanus Folk stars with be over the next five years.

This tool as presented in Figure 9 has two objectives. The first one is to display the careers of a selection of artists. To achieve this, multiple visualizations work together to show the connections between the artists and their releases, influences, achievements, popularity, genres, activity, and roles. The second one is to compare a selection of artists based on the number of releases, years of activity, and notability. The aim in this comparison is to estimate the next rising star for a given time and genre.

To begin using the tool, a set of artists have to be selected. The tool has a selection of filters to apply, such as number of releases (song/album), genre, role, and years of activity. These filters directly affect the table, where artists can be selected. To provide insights about the dataset, two histograms for song and album releases among the entire dataset are presented. These histograms show that around 90% of the entire dataset does not have more than 1 song or album release. Consequently, both filters are set to 1, reducing the sample from 11,579 to 1,449 artists.

With the applied filters, 3 artists are selected from the table. Jay Walters was selected since he has the highest number of notable releases (37). Kimberly Snyder was selected since she has the highest number of notable releases as a performer (20), and Sailor Shift is selected with the highest number of releases (46) among the dataset. The selection is made by toggling the boxes, and using the UI button to pass the list to the career analysis section.

Figure 9: The visual analytics tool for question 3.

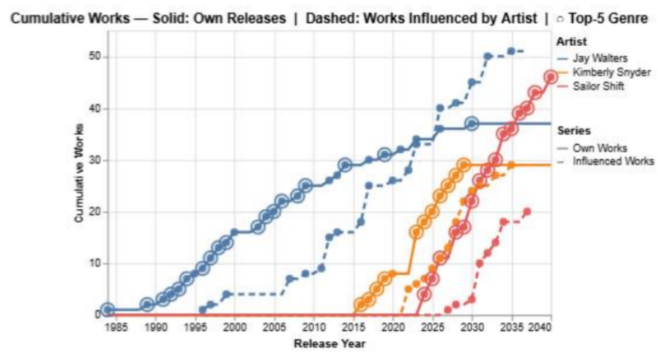

Figure 10: Cumulative works line chart: solid lines show own releases, dashed lines show works influenced by each artist. Circles mark releases in the top 5 notable genres.

The first graph as presented in Figure 10 is a line chart with year on the horizontal and cumulative counts on the vertical. The solid lines show the cumulative number of releases, and the dashed lines show the cumulative number of influences. On the solid line, circles appear on nodes if the artist released a song or album among the top 5 most notable genres of that year. The graph captures career highlights such as rise in popularity, breaks, and release frequencies. The graph shows that Jay Walters had a slow but steady release pace whereas Kimberly Snyder and Sailor Shift were much more aggressive with their releases, with Sailor Shift continuing at the same pace today. Although having 32 years between their debuts, both Jay Walters and Kimberly Snyder can be assumed to quit music at 2028, the same year Sailor Shift rose in popularity.

The second graph as presented in Figure 11 the same vertical is a line chart showing the influence ratio of the artists, gathered from dividing cumulative influences to cumulative releases, in a cohort year view. This view

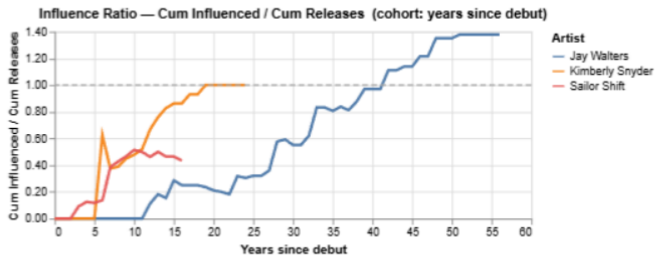

Figure 11: Influence ratio (cumulative influences / cumulative releases) in cohort-year view. The dashed horizontal line marks the breakeven point at $y = 1$.

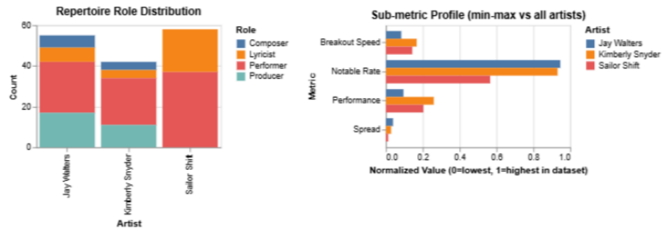

Figure 13: Repertoire role distribution (left) and sub-metric profile with min-max normalisation across the filtered cohort (right).

enables the user to better compare the artists as if their debut was the same year. A horizontal line is inserted on $y = 1$ as a breakeven line. The graph shows that it took Jay Walters 10 years to have his first influence on somebody else's release, and 40 years to have more influences than his releases. It took Kimberly Snyder 5 years to influence someone else's release, and 18 years to have her influences reach her releases. For Sailor, it only took her 2 years to influence. Although she had a similar pace as the other two, for the last 5 years the rate of influences is declining. A reason for this might be that Sailor's influences from the dataset were mostly pointing to groups and persons, except for one album, whereas Kimberly and Jay had only influenced songs and albums. Therefore, their influences are not the same quantitatively.

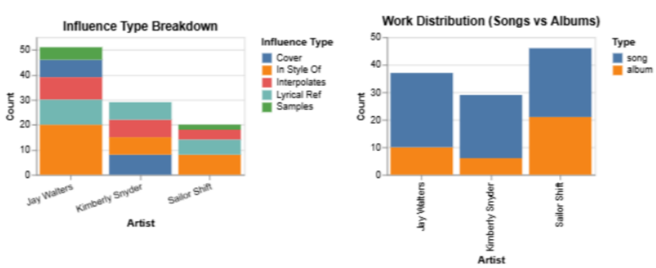

Figure 12: Influence type breakdown (left) and work distribution — songs vs. albums (right) for the three selected artists.

The third section contains 6 bar charts to illustrate distributions among different areas:

The first graph as presented in Figure 12 is a breakdown of influences, showing in which ways did the artists influenced others. Besides sampling Kimberly Snyder and covering Sailor Shift, all artists made influences in the areas available (sampling, lyrical reference, style, interpolation, cover). Although the distributions look similar, Jay Walters has had a heavy influence on style.

The second graph as presented in Figure 12 shows the distribution of work, either being songs or albums. It shows that among the three, Shift took more part in releasing albums, accounting for almost half of his discography.

The third graph as presented in Figure 13 is a breakdown of roles on releases. The graph shows that throughout the discographies of the artists, Shift had only been a performer and a lyricist whereas Snyder and Walters roles also as composer and producers. However, the most common role for all three artists was the performer role.

The fourth graph as presented in Figure 13 shows the positions of the three artist about some parameters, among the sample that is filtered by the minimum song and album UI slider. To show the parameters on the same scale, a min-max normalization among the sample is applied. Moreover, the normalization also acts as a bridge between the artist set and the dataset. The minimum song and album filters play a crucial role in eliminating anomalies that do not bring significant arguments to the table.

- The first parameter is **breakout speed** and is found by dividing the debut year to the first influence year. The results show that Shift and Snyder have similar breakout speeds surpassing Waters.
- The second parameter is **notable rate** and is found by dividing the cumulative notable releases to cumulative releases. The results show that Walters and Snyder have around 90% notable rates whereas Shift has around 55%.
- The third parameter is **performance** and is found by dividing the number of notable releases to the years of activity, which is calculated by taking the difference between last and first releases. The results show that Snyder managed to make the most of her time, closely followed by Snyder. This is in agreement with the release frequency findings from the first line chart.
- The final parameter is **spread** and is found by dividing total influences to total releases. All three artists have similar spread. The effect of normalization is heavily observed here with all three artists valued close to 0, since there are artists in the dataset with a single release and 150+ influences.

The last two graphs as presented in Figure 14 are genre distributions of the artists where one of them shows the genres the artists are influenced by and the other shows

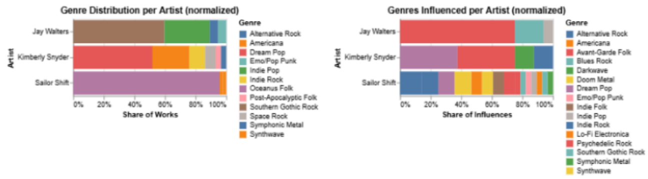

Figure 14: Genre distribution of own releases (left) and genres drawn from in influenced works (right), normalised per artist.

the genre distributions of their releases. The graph shows that Shift remained loyal to her genre, “Oceanus Folk” whereas the others tried 4–6 different genres. However, when the influence distributions are observed, it is seen that Shift was influenced by 16 different genres in her work while the other artists are only influenced by at most 4 different genres. This is a great indicator of how successful artists with many notable releases can differ on how open they are to try different genres or get influenced by.

The comparisons between these successful artists showed that making decisions based on influences might be misleading. The data types may not always be comparable, and there is a delay in the acceleration of influences, possibly related to popularity effect. Therefore, when trying to estimate the new rising stars, notable releases are expected to be the deciding factor. Finally, the analysis tool managed to clearly illustrate the differences in careers of three successful artists, who also happen to have the top 3 most notable releases among the whole dataset.

Figure 15: Filters for the artist set to estimate the next rising star.

To make the estimations, the datasets are filtered once more, as presented in Figure 15 setting genre to “Oceanus Folk”, latest activity higher than 2035 to make prediction for today (2040) and discard those who have been inactive. The filtering results with 24 artists and groups. To make the analysis, the UI button “Pass list to genre selector” is used to pass the rows to the scatterplot.

The scatterplot is presented in Figure 16. In addition to the x and y values which are years of activity and notable release rate, the scatter plot utilizes size, and shape to show artists of the same groups, and number of releases. When we look at the x axis, we can see that there are 4 new groups that are younger than Ivy Echoes, which are stationed in the middle of the horizontal. Moreover, almost all groups have around the same notable rate of 50%, except Copper Canyon Ghosts, who have only notable releases. In addition to that, in the volume of releases, there is a clear distinction between Ivy Echoes and the rest of the groups, which is expected.

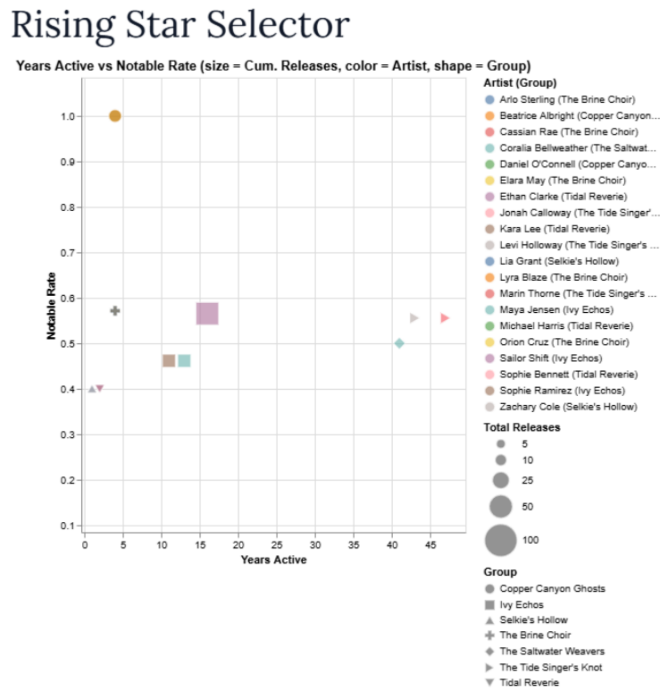

Figure 16: Rising Star Selector scatterplot: years active vs. notable rate, with size encoding total releases and shape encoding group membership.

Therefore, based on the scatterplot, two estimations for the next rising stars of ‘Oceanus Folk’ can be made clearly. These will be the **Copper Canyon Ghosts** and **The Brine Choir**. The last prediction requires further analysis since it was observed that the graph could not conclude a distinction between Tidal Reverie and Selkie’s Hollow. Therefore, the career analysis tool will be used once more to compare more in depth.

Based on the career analysis tool, it was observed that Selkie’s Hollow has been more recently active, and has influences from 4 different genres whereas the only inspiration of Tidal Reverie was Indie Rock. Taking into account how Sailor was inspired by many different genres as an artist of ‘Oceanus Folk’, the last prediction for the next rising star of the coming years will be **Selkie’s Hollow**, along with Copper Canyon Ghosts and The Brine Choir.

Discussion & Conclusions

In this report, three different visual analytics tools were developed to tackle three different questions. With the tool developed, a user can understand the influences and collaborations of “Sailor Shift”, with added depth on understanding direct and indirect influences and genre distributions. Moreover, a user can dive deep on the “Oceanus Folk” genre, identifying spread, genres influenced by, and top representatives, in any given year range. Finally, a user can select any artist set they want using filters, and a table view to analyze and contrast their careers and compare a much larger artist set using a scatterplot to identify which artists have a more promising future.

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Appendix 1: Division of Tasks

- **Meghan:** Literature Review & Question 1
- **Christiana:** Introduction & Question 2
- **Kaan:** Discussion/Conclusion & Question 3

Appendix 2: AI Usage Log

Question 1

Claude was used at several points during this project to support code review, auditing, and debugging — always under my direction, and never as a substitute for my own analytical judgement. For code review and auditing, I directed Claude to read through my analysis pipeline and flag issues relating to data provenance, redundant computations, and inconsistencies in how variables were named and passed between processing stages; any issues it identified were reviewed by me before anything was changed, and I made the final call on what to fix. On the debugging and optimization side, I used Claude to help trace the root cause of unexpected outputs at specific points in the pipeline — particularly where data transformations were producing results that didn't match my expectations — and to suggest targeted improvements to how certain intermediate datasets were being constructed. Throughout, all decisions about methodology, analytical scope, and visualization design were my own. A full log of prompts used throughout this project has been saved to a separate document and is available upon request. Examples of prompts used are included below:

1. *Prompt:* Asked how to best structure the influence data across multiple dataframes so it would be easiest to use for visualisation analysis in Marimo and Altair, given the range of possible edge interaction types in the graph.

   - *Response:* Recommended building three flat tables — one raw edge-level log, one artist-resolved influence table, and one network node table — following Altair's requirement of one observation per row, and explained why the enriched list-style tables built earlier were not suitable for charting.

2. *Prompt:* Asked whether network nodes for Q2 should represent artists or works, given that the raw influence edges in the data connect songs and albums rather than people directly.

   - *Response:* Recommended using artists as nodes rather than works, on the basis that the research questions were framed around people and that work nodes would create unnecessary density; explained that the pipeline's artist-resolution step already provided the correct level of abstraction.

3. *Prompt:* Asked why pre-2023 activity was appearing for Sailor-connected artists in the Q3 visualisation, and how it could be corrected.

   - *Response:* Explained that the `influenced_by_sailor` flag was applied statically across all years of an artist's output regardless of when the influence relationship formed; a temporal guard enforcing `release_year < 2023` → `'No influence'` was the appropriate correction.

4. *Prompt:* Directed a read-only audit of the full analysis pipeline from `find_sailor` onwards, covering data provenance, redundancy, function correctness, and Marimo cell consistency.

- *Response:* Returned a structured audit report categorised into redundancy issues (R1–R8) and consistency issues (C1–C3), describing each problem and its location without making any changes.

5. *Prompt:* After the audit identified that two separate pipeline functions were each independently constructing the same work-to-artist lookup table from the same source data, asked for the redundancy to be resolved.

   - *Response:* Identified that the lookup computed in the earlier function could be returned as a shared output and passed directly into the later function as a parameter, eliminating the duplicate construction without changing the data or behaviour of either function.

Question 2

For Question 2, ChatGPT was used as a support tool to assist with the development of the visual analytics dashboard, which includes three charts: the time trend, heatmap, and lollipop chart. Its role was primarily technical and advisory, assisting with coding, troubleshooting errors and recommending how to improve the aesthetic layout of the final dashboard. ChatGPT provided guidance on data preprocessing techniques, graph design choices, and debugging issues related to variable definitions and execution order. All suggestions and outputs generated by AI were critically evaluated and verified by me, before being implemented in practice. It was not used as a substitute for independent thinking or decision making, but rather as a supportive source that enhances efficiency (e.g. spots errors when I might have overlooked them) and problem solving, throughout the process of developing this assignment.

Purpose of Use:

- Help with data preprocessing methods
- Debugging the attempt to integrate multiple graphs into one dashboard
- Debugging and refining code within the Marimo notebook (multiple errors from variable definitions)

Example Prompts and Outputs:

1. "What is an appropriate method in pandas to aggregate temporal data by year, in order to analyze trends over time?"

   - *Output:* Suggested using the `groupby` approach to construct a yearly aggregation of influenced songs.

2. "How does variable redefinition and execution order affect results in Marimo? I am getting multiple errors, even though my code itself is correct."

   - *Output:* Suggested sharing code cells, to detect any repetition of variable definitions and spotted the problem that was missed.

3. "What are recommended alternatives for bar charts that display top performing artists, that appeal more to the audience?"

   - *Output:* Suggested different types of alternatives, and we decided to implement the lollipop chart, as it reduced visual clutter.

4. "What is a good choice for color scales in a heatmap, to show magnitude but maintain readability?"

   - *Output:* Suggested a sequential color scale (`"tealblues"`), to represent the influence counts.

5. "What is the best way to integrate my 3 charts in Marimo into one final dashboard?"

   - *Output:* Suggested the function `mo.vstack()` to stack them into a readable sequence vertically.

Question 3

Claude was used for both the preparation of data from JSON to dataframes, and creating visualizations for question three.

For the first part, Claude was prompted along with the data description file as:

> *"Analyse the file MC1 Data Description file. This file contains the relations between different nodes, and edge names. Based on the relationships, use MC1 graph.json to create dataframes for each base node group, and create their corresponding edges where the base node is in the destination. Before proceeding, write down all the relationships you analysed, so I can see if you are missing any dependency."*

After this prompt, the dataframes are prepared for the questions, and each of them are checked if there are any missing information.

For the third part, Claude was utilized in two different ways. First, I prompted and navigated Claude to create the tables that I need for the analysis. One of these tables is an influence table that contains all influence relations for types such as song, album, person and group. The other table is a career table that contains distinct artist IDs and contains career information about influences, releases, activities, genre and so on. The last two tables were artist × work and artist × influence tables. After preparing all the information I needed, I health checked the information on the tables using direct node connections from the `.json`, and started creating visualizations. After many iterations on visualizations and further health checks, I prompted Claude again to combine them to a dashboard using the tutorial from week 5 as reference, including my desired filters, buttons, and dependencies. The initial prompt can be found below:

> *Act as a data visualization expert. Using the file as reference, create me a dashboard with the following elements:*
>
> 1. *Filters: Minimum number of songs, minimum number of albums, genre, role (performer, composer, lyricist, producer), release type (song, album), minimum/maximum years of activity, range of latest release year (all these were already columns of the career table, and were already prepared separately).*
> 2. *Two histograms for song and album distributions, static, with a tooltip, made previously.*
> 3. *One artist per row table containing career information, with tick boxes on the first columns that stores the unique IDs, refer back to the career table.*
> 4. *A UI element that is clickable that feeds the stored artist IDs to the next visualization, consecutive use of this button does not overwrite but adds new artist IDs, unless the clear button from the next section is used.*
> 5. *A Career Dashboard, containing the line charts and the bar plots made previously, with a UI element to clear the inputs of the dashboard.*
> 6. *A UI element that is clickable that feeds the entire filtered dataset loaded on the table to the next section.*
> 7. *A Rising Star Dashboard, containing the scatter plot made previously.*

Claude got everything set up easily because most of them were already done, but as a visual analytics tool, there were iterations on the positioning of the tools and update relationships.